Product Reference Manual



Description

The ARIES v3.0 is a fully indigenous and a “Made in India” product to get started with basic microprocessor programming and embedded systems. This board is built upon a RISC-V ISA compliant VEGA Processor with easy-to-use hardware and software. The VEGA SDK also provides full ecosystem with numerous examples and support documentation. This board is designed and developed by Centre for Development of Advanced Computing (C-DAC) as part of the Digital India RISC-V (DIR-V) Program, by the Ministry of Electronics and Information Technology, Government of India.

Target areas/Applications

Low power IoT, Sensor fusion, Smart Meter, System supervisors, Remote sensors, Wearable devices, Toy and electronic education equipment, Legacy 8/16-bit applications, Industrial networking and many more…

Features

Controller	THEJAS32
SRAM	256KB
Flash	2MB
Input voltage	7-12V
PWM pins	8nos
Analog Input pins	4nos
SPI	3nos
UART	2nos
I2C	2nos
GPIO	23nos
RGB	1nos (3x GPIO)
DC Current per I/O pin	12mA
IO Voltage	3.3V
Clock speed	100MHz
Length	69mm
Width	53mm

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1. The Board

ARIES v3.0 is a development platform based on THEAJS32 ASIC which operates at a frequency of 100MHz. THEJAS32 SoC includes VEGA ET1031 Microprocessor, 256KB internal SRAM, Three UARTs, Four SPIs, Three TIMERs, Eight PWMs, Three I2C interfaces, 32 GPIOs etc. This board contains everything needed to support standalone operation. To get started simply connect the board to a computer with a Micro-C USB Cable or power it with a AC/DC adapter or a battery.

2. Handling the Board

To avoid causing any damage or malfunctions; it is important to be mindful of the following points when handling or operating the board:

- To prevent any damage make sure to handle the board while taking electrostatic discharge (ESD) precautions.
- Power down the board by disconnecting the board from USB port.

2.1 Recommended Operating Conditions

Symbol	Description	Min	Max
	Conservative thermal limits for the whole board:	-0 °C (100°F)	85 °C (185°F)

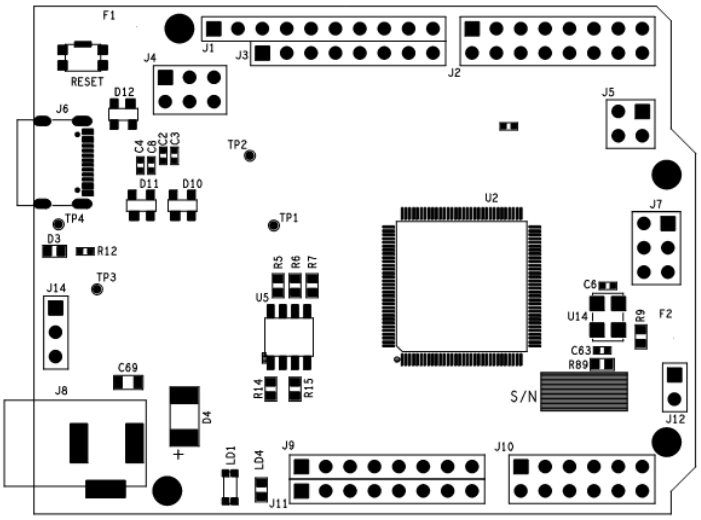
2.2 Power Consumption

Symbol	Description	Min	Typ	Max	Unit
VINMax	Maximum input voltage from VIN pad	7	-	12	V
VUSBMax	Maximum input voltage from USB connector	-	5.5	V	
PMax	DC Current per I/O Pin	-	-	12	mA

3. Functional Overview

3.1 Board Topology

Top view



Ref.	Description	Ref.	Description
J8	Power jack	U2	THEJAS32 SoC
J6	USB C Connector	U1	3.3V Regulator
J12	Boot Select Header	U4	100 MHz Oscillator
RESET	Reset Button	U5	SPI Boot Flash Memory
LD1	RGB LED	U6	Op-amp
LD4	Processor HeartBeat LED	U7	1.2V Regulator

3.2 Processor

The main controller is THEAJS32 SoC which operates at a frequency of 100MHz. It includes VEGA ET1031 Microprocessor, 256KB internal SRAM, Three UARTs, Four SPIs, Three TIMERs, Eight PWMs, Three I2C interface, 32 GPIOs. Most of its pins are connected to the external headers, however some are reserved for internal communication.

3.3 THEJAS32 pinout

Pin #	Pin Name	Pin Description	Type
1	GPIO19	General purpose IO GPIO1 (3).	I/O
2	GPIO18	General purpose IO GPIO1 (2).	I/O
3	VSSIO	Ground reference for IO pins.	S
4	VDDIO	Positive supply for IO pins. Connect to 3.3V supply.	S
5	GPIO17	General purpose IO GPIO1 (1).	I/O
6	GPIO16	General purpose IO GPIO1 (0).	I/O
7	SPI_MOSI3	SPI 3 Master Out Slave In.	O
8	VDD	Positive supply for logic. Connect to 1.2V supply.	S
9	VSS	Ground reference for logic.	S
10	SPI_MISO3	SPI 3 Master In Slave Out.	I
11	SPI_SCLK3	SPI 3 Clock.	O
12	SPI_SS3	SPI 3 Chip Select.	O
13	VSSIO	Ground reference for IO pins.	S
14	VDDIO	Positive supply for IO pins. Connect to 3.3V supply.	S
15	BOOT_SEL	Boot select.	I
16	PROC_HB	Heartbeat signal.	O
17	RFIU1	Connect to GND.	NA
18	VDD	Positive supply for logic. Connect to 1.2V supply.	S
19	VSS	Ground reference for logic.	S
20	RFIU2	Connect to GND through a 1K resistor.	NA
21	RFIU3	JTAG TDO. Left unconnected.	NA
22	RFIU4	JTAG TMS. Connect to GND through a 1K resistor.	NA
23	RFIU5	JTAG TDI. Connect to GND through a 1K resistor.	NA
24	VSSIO	Ground reference for IO pins.	S
25	VDDIO	Positive supply for IO pins. Connect to 3.3V supply.	S
26	VDD	Positive supply for logic. Connect to 1.2V supply.	S
27	VSS	Ground reference for logic.	S

28	RFIU6	JTAG TCK. Connect to GND through a 1K resistor	NA
29	RFIU7	JTAG TRST. Connect to GND through a 1K resistor	NA
30	RFIU8	Test mode select. Connect to GND through a 1K resistor.	NA
31	I2C_SDA2	I2C 2 Serial Data.	I/O
32	I2C_SCL2	I2C 2 Serial Clock.	I/O
33	I2C_SCL0	I2C 0 Serial Clock.	I/O
34	I2C_SDA0	I2C 0 Serial Data.	I/O
35	VSS	Ground reference for logic.	S
36	VDD	Positive supply for logic. Connect to 1.2V supply.	S
37	VDDIO	Positive supply for IO pins. Connect to 3.3V supply.	S
38	VSSIO	Ground reference for IO pins.	S
39	SPI_SS1	SPI 1 Chip Select.	O
40	SPI_SCLK1	SPI 1 Clock.	O
41	SPI_MISO1	SPI 1 Master In Slave Out.	I
42	SPI_MOSI1	SPI 1 Master Out Slave In.	O
43	PUSH_RESETN	Reset. (ACTIVE LOW)	I
44	CLK	System Clock.	I
45	UART_TX1	UART 1 Serial Out / Transmit.	O
46	VDDIO	Positive supply for IO pins. Connect to 3.3V supply.	S
47	VSSIO	Ground reference for IO pins.	S
48	VSS	Ground reference for logic.	S
49	VDD	Positive supply for logic. Connect to 1.2V supply.	S
50	UART_RX1	UART 1 Serial In / Receive.	I
51	GPIO15	General purpose IO GPIO0 (15).	I/O
52	GPIO14	General purpose IO GPIO0 (14).	I/O
53	GPIO13	General purpose IO GPIO0 (13).	I/O
54	GPIO12	General purpose IO GPIO0 (12).	I/O
55	GPIO11	General purpose IO GPIO0 (11).	I/O

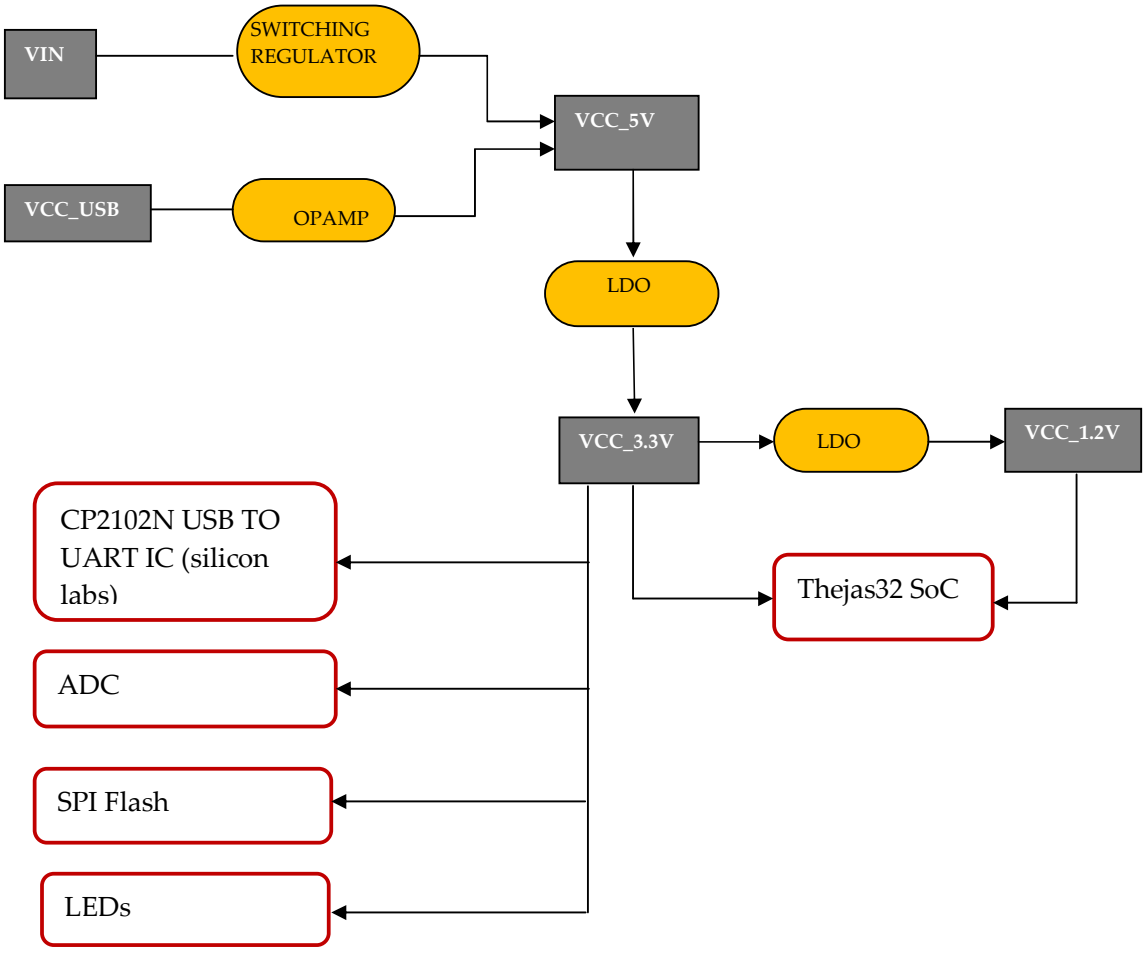
56	VSS	Ground reference for logic.	S
57	VDD	Positive supply for logic. Connect to 1.2V supply.	S
58	GPIO10	General purpose IO GPIO0 (10).	I/O
59	VDDIO	Positive supply for IO pins. Connect to 3.3V supply.	S
60	VSSIO	Ground reference for IO pins.	S
61	GPIO9	General purpose IO GPIO0 (9).	I/O
62	GPIO8	General purpose IO GPIO0 (8).	I/O
63	GPIO7	General purpose IO GPIO0 (7).	I/O
64	GPIO6	General purpose IO GPIO0 (6).	I/O
65	GPIO5	General purpose IO GPIO0 (5).	I/O
66	GPIO4	General purpose IO GPIO0 (4).	I/O
67	VSS	Ground reference for logic.	S
68	VDD	Positive supply for logic. Connect to 1.2V supply.	S
69	VDDIO	Positive supply for IO pins. Connect to 3.3V supply.	S
70	VSSIO	Ground reference for IO pins.	S
71	GPIO3	General purpose IO GPIO0 (3).	I/O
72	GPIO2	General purpose IO GPIO0 (2).	I/O
73	GPIO1	General purpose IO GPIO0 (1).	I/O
74	GPIO0	General purpose IO GPIO0 (0).	I/O
75	PWM_7	Pulse Width Modulation.	O
76	PWM_6	Pulse Width Modulation.	O
77	PWM_5	Pulse Width Modulation.	O
78	VSS	Ground reference for logic.	S
79	VDD	Positive supply for logic. Connect to 1.2V supply.	S
80	PWM_4	Pulse Width Modulation.	O
81	PWM_3	Pulse Width Modulation.	O
82	PWM_2	Pulse Width Modulation.	O
83	VDDIO	Positive supply for IO pins. Connect to 3.3V supply.	S
84	VSSIO	Ground reference for IO pins.	S
85	PWM_1	Pulse Width Modulation.	O

86	PWM_0	Pulse Width Modulation.	O
87	SPI_MOSI0	SPI 0 Master Out Slave In.	O
88	VSS	Ground reference for logic.	S
89	VDD	Positive supply for logic. Connect to 1.2V supply.	S
90	SPI_MISO0	SPI 0 Master In Slave Out.	I
91	SPI_SCLK0	SPI 0 Clock.	O
92	SPI_SS0	SPI 0 Chip Select.	O
93	VDDIO	Positive supply for IO pins. Connect to 3.3V supply.	S
94	VSSIO	Ground reference for IO pins.	S
95	I2C_SDA1	I2C 1 Serial Data.	I/O
96	I2C_SCL1	I2C 1 Serial Clock.	I/O
97	SPI_MOSI2	SPI 2 Master Out Slave In.	O
98	SPI _MISO2	SPI 2 Master In Slave Out.	I
99	VDD	Positive supply for logic. Connect to 1.2V supply.	S
100	VSS	Ground reference for logic.	S
101	SPI_SCLK2	SPI 2 Clock.	O
102	SPI_SS2	SPI 2 Chip Select.	O
103	VSSIO	Ground reference for IO pins.	S
104	VDDIO	Positive supply for IO pins. Connect to 3.3V supply.	S
105	UART_RX2	UART 2 Serial In / Receive.	I
106	UART_TX2	UART 2 Serial Out / Transmit.	O
107	UART_RX0	UART 0 Serial In / Receive.	I
108	UART_TX0	UART 0 Serial Out / Transmit.	O
109	GPIO31	General purpose IO GPIO1 (15).	I/O
110	GPIO30	General purpose IO GPIO1 (14).	I/O
111	GPIO29	General purpose IO GPIO1 (13).	I/O
112	VDD	Positive supply for logic. Connect to 1.2V supply.	S
113	VSS	Ground reference for logic.	S
114	VSSIO	Ground reference for IO pins.	S
115	VDDIO	Positive supply for IO pins. Connect to 3.3V supply.	S

116	GPIO28	General purpose IO GPIO1 (12).	I/O
117	GPIO27	General purpose IO GPIO1 (11).	I/O
118	GPIO26	General purpose IO GPIO1 (10).	I/O
119	GPIO25	General purpose IO GPIO1 (9).	I/O
120	GPIO24	General purpose IO GPIO1 (8).	I/O
121	GPIO23	General purpose IO GPIO1 (7).	I/O
122	GPIO22	General purpose IO GPIO1 (6).	I/O
123	VSSIO	Ground reference for IO pins.	S
124	VDDIO	Positive supply for IO pins. Connect to 3.3V supply.	S
125	VDD	Positive supply for logic. Connect to 1.2V supply.	S
126	VSS	Ground reference for logic.	S
127	GPIO21	General purpose IO GPIO1 (5).	I/O
128	GPIO20	General purpose IO GPIO1 (4).	I/O

S-Supply, I/O - Input/ Output, I - Input, O - Output, RFIU - Reserved for internal use

3.4 Power tree

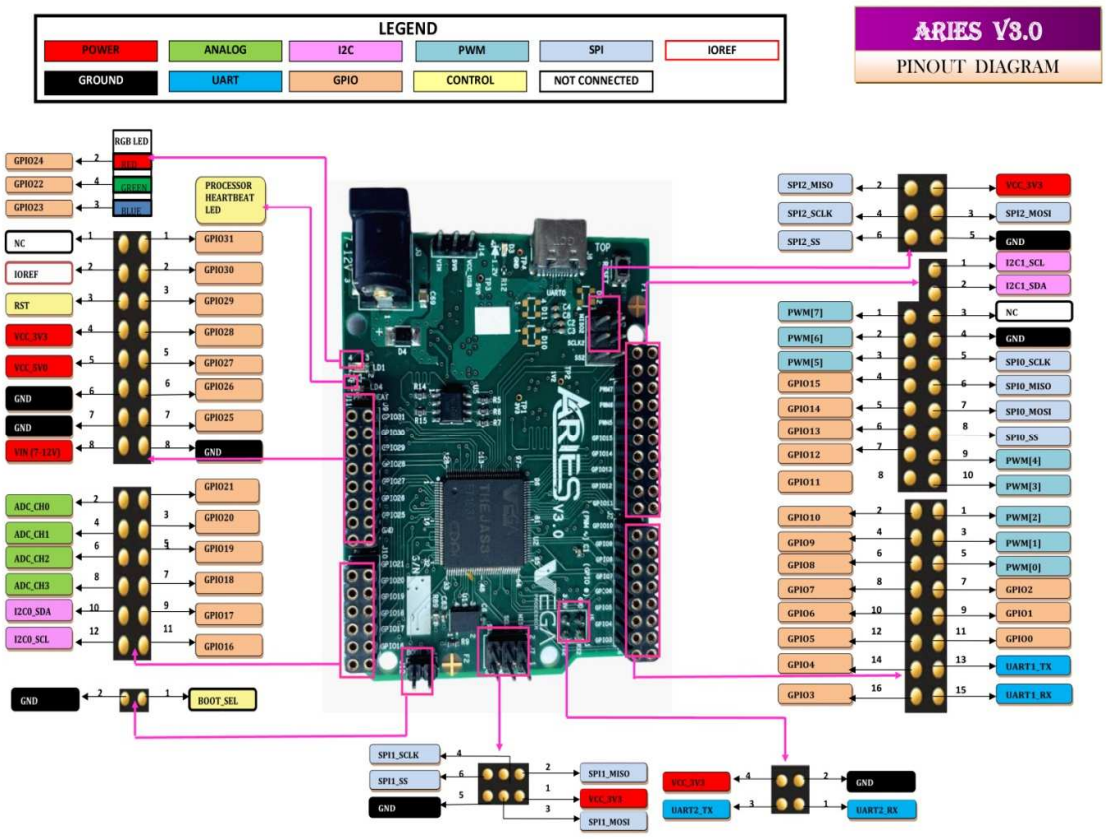


4. Board Operation

4.1 Getting Started

- To use Vega Arduino IDE for programming follow the steps given in the link below
  - For Linux; https://bit.ly/vega-linux
  - For Windows; https://bit.ly/vega-windows
- To use Eclipse IDE for programming follow the steps given in the link below
  - https://cdac-vega.gitlab.io/sdkuserguide.html

5. Connector Pinout



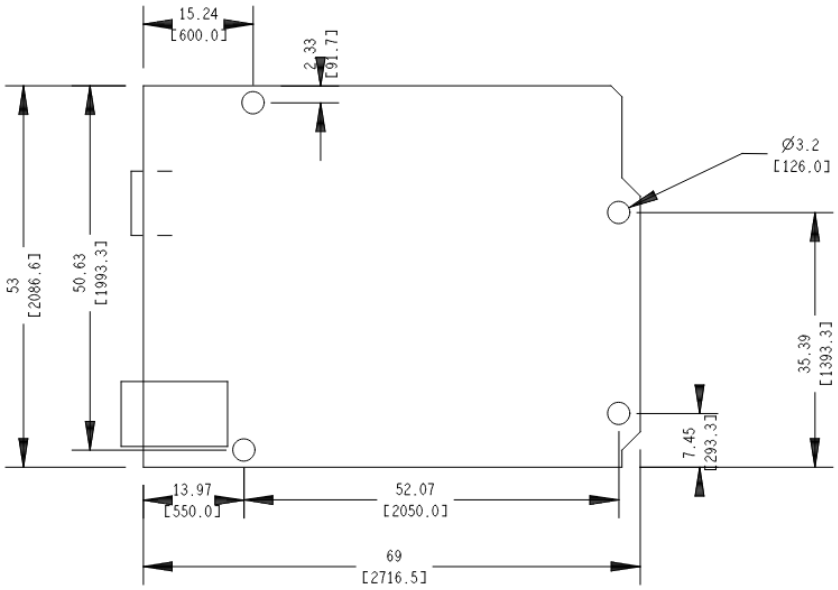
SL No.	SPECIFICATIONS	THEJAS 32	ARIES BOARD	REMARKS
1	SPI (4)	SPI_SS0	J1_8	Connected to HEADER J1
2		SPI_SCLK0	J1_5	
3		SPI_MISO0	J1_6	
4		SPI_MOSI0	J1_7	
5		SPI_SS1	J7_6	Connected to HEADER J7
6		SPI_SCLK1	J7_4	
7		SPI_MISO1	J7_2	
8		SPI_MOSI1	J7_3	
9		SPI_SS2	J4_6	Connected to HEADER J4
10		SPI_SCLK2	J4_4	
11		SPI_MISO2	J4_2	
12		SPI_MOSI2	J4_3	
13		SPI_SS3	U5_1	Connected to SPI BOOT FLASH IC
14		SPI_SCLK3	U5_6	
15		SPI_MISO3	U5_2	
16		SPI_MOSI3	U5_5	
17	I2C (3)	SCL0	J10_12	Connected to HEADER J10
18		SDA0	J10_10	
19		SCL_1	J1_1	Connected to HEADER J1
20		SDA_1	J1_2	
21		SCL_2	U9_10	Connected to I2C ADC IC
22		SDA_2	U9_9	
23	UART (3)	UART_RXD0	U10_18	Connected to UART TO USB INTERFACE IC
24		UART_TXD0	U10_17	

25		UART_RX1	J2_15	Connected to HEADER J2
26		UART_TX1	J2_13	
27		UART_RX2	J5_1	Connected to HEADER J5
28		UART_TX2	J5_2	
29	PWM (8)	PWM_0	J2_5	Connected to HEADER J2
30		PWM_1	J2_3	
31		PWM_2	J2_1	
32		PWM_3	J1_10	Connected to HEADER J1
33		PWM_4	J1_9	
34		PWM_5	J3_3	Connected to HEADER J3
35		PWM_6	J3_2	
36		PWM_7	J3_1	
37	GPIO (32)	GPIO0	J2_11	Connected to HEADER J2
38		GPIO1	J2_9	
39		GPIO2	J2_7	
40		GPIO3	J2_16	
41		GPIO4	J2_14	
42		GPIO5	J2_12	
43		GPIO6	J2_10	
44		GPIO7	J2_8	
45		GPIO8	J2_6	
46		GPIO9	J2_4	
47		GPIO10	J2_2	
48		GPIO11	J3_8	Connected to HEADER J3
49		GPIO12	J3_7	
50		GPIO13	J3_6	
51		GPIO14	J3_5	

52		GPIO15	J3_4	
53		GPIO16	J10_11	Connected to HEADER J10
54		GPIO17	J10_9	
55		GPIO18	J10_7	
56		GPIO19	J10_5	
57		GPIO20	J10_3	
58		GPIO21	J10_1	
59		GPIO22	LD1_4	Connected to GREEN of RGB LED
60		GPIO23	LD1_3	Connected to BLUE of RGB LED
61		GPIO24	LD1_2	Connected to RED of RGB LED
62		GPIO25	J9_7	Connected to HEADER J9
63		GPIO26	J9_6	
64		GPIO27	J9_5	
65		GPIO28	J9_4	
66		GPIO29	J9_3	
67		GPIO30	J9_2	
68		GPIO31	J9_1	
69	CLOCK	CLK_100M	U4_3	Connected to OSCILLATOR
70	RESET	PUSH_RESETN	U 8_1	Connected to RESET IC
71	HEART BEAT LED	PROC_HEART_BEAT	LD4_1	Connected to LED
72	BOOT SELECT	BOOT_SEL	J12_1	Connected to HEADER J12

73	ADC_CH0	_	J10_2,U9_4	ANALOG INPUTS Connected to HEADER J10 & I2C ADC IC
74	ADC_CH1	_	J10_4,U9_5	
75	ADC_CH2	_	J10_6,U9_6	
76	ADC_CH3	_	J10_8,U9_7	

5.1 Board Outline & Mounting Holes



(Dimensions in mm [mil])

6. Company Information

Company name	**C-DAC**
Company Address	Hardware Design Group Centre for Development of Advanced Computing (C-DAC) Thiruvananthapuram, Kerala – 695033 Fax: 0471-2723456 Email: vega@cdac.in www.vegaprocessors.in www.cdac.in

7. Reference Documentation

Reference	**Link**
ARIESv3.0 details	https://vegaprocessors.in/devboards/ariesv3.html